

MATH 350-2 Advanced Calculus

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Outline

- 1 Real Analysis Lecture 18
 - Connectedness

Warm-up

Let (S, d) be a metric space. Prove that if $f : S \rightarrow \mathbb{R}$ and $g : S \rightarrow \mathbb{R}$ are continuous at $a \in S$, then so is $f + g$.

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Connected spaces

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- the real line with the discrete metric $(\mathbb{R}, d_{disc})$ is connected
- euclidean space $\mathbb{R}^n$ is connected
- an interval $I \subseteq \mathbb{R}$ with the Euclidean metric is connected

Challenge!

Problem

Prove that the set $\mathbb{Q} \subseteq \mathbb{R}$ is disconnected.

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- $U \cap V = \emptyset$
- U, V are open
- $U \cup V = \mathbb{R}$

In particular, U is a nonempty, proper subset of $\mathbb{R}$ which is both open and closed. □

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By the Representation Theorem for Open Sets:

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$$U = (a_1, b_1) \cup (a_2, b_2) \cup \dots$$

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is a union of disjoint open intervals.

In particular, b_1 is an accumulation point of U but not an element of U .

This contradicts the fact that U is closed. □

Images of connected sets

Theorem

Let (S, d_S) and (T, d_T) be metric spaces and $f : S \rightarrow T$ be a function.

If f is continuous and S is connected, then $f(S)$ is connected.

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Suppose $f(S)$ is disconnected.

Then $f(S) = U \cup V$ for nonempty, disjoint, open sets
 $U, V \subseteq f(S)$.

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Proof.

Suppose $f(S)$ is disconnected.

Then $f(S) = U \cup V$ for nonempty, disjoint, open sets $U, V \subseteq f(S)$.

Then $f^{-1}(U)$ and $f^{-1}(V)$ are non-empty, disjoint, open sets and

$$f^{-1}(U) \cup f^{-1}(V) = f^{-1}(U \cup V) = f^{-1}(f(S)) = S.$$

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This is a contradiction. □

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We can make any interval as a continuous image:

$$f : \mathbb{R} \rightarrow \mathbb{R}$$
$$f(x) = \frac{b+a}{2} + \frac{(b-a)x}{1+x^2}$$

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We can make any interval as a continuous image:

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Intermediate value theorem

Theorem (Intermediate Value Theorem)

*Suppose $f : [a, b] \rightarrow \mathbb{R}$ is a continuous function.
Then for any value y between $f(a)$ and $f(b)$, there exists
 $c \in [a, b]$ with $f(c) = y$.*

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Without loss of generality, assume $f(a) < f(b)$.

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$$U = (-\infty, y) \cap J \quad \text{and} \quad V = (y, \infty) \cap J$$

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These are nonempty, disjoint, and open in the subspace J of $\mathbb{R}$.

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This is a contradiction! □